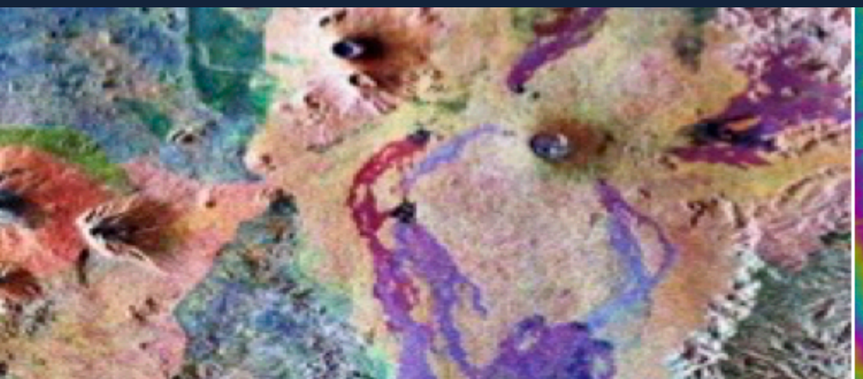
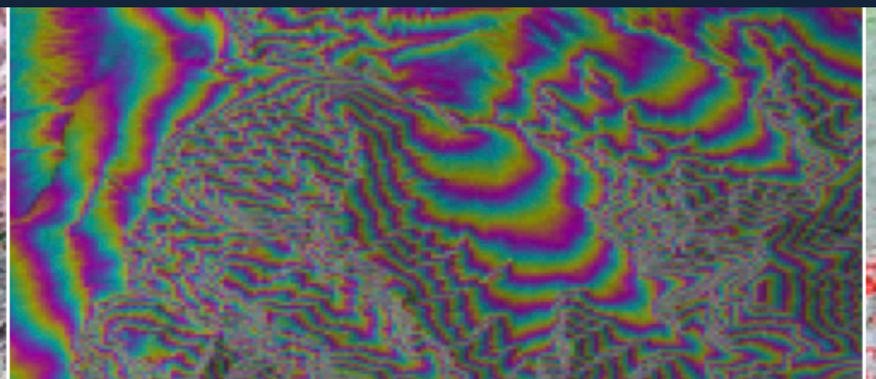
ENVIRONMENTAL GEOHAZARDS ENGINEERING · INTRODUCTION

SAR and InSAR for Geohazards

Radar satellites, ground motion and engineering

AKSHAY RAJ MANOCHA







WHY THIS MATTERS

The ground is always moving. Landslides creep, cities sink, faults slip and volcanoes swell, often by only millimetres a year.

Satellite radar interferometry (InSAR) measures that motion from space: over thousands of square kilometres at once, day or night, through cloud.

THIS LECTURE

Outline

01 Remote sensing and radar basics
Passive and active sensors, radar bands, and fifty years of radar satellites

02 What is SAR?
How a radar satellite takes a picture, what the picture shows, and its distortions

03 What is InSAR?
Phase, fringes, line of sight, time series and coherence

04 Satellites and orbits
How radar satellites circle the Earth, Sentinel-1, NISAR and other missions

05 Hazards and engineering
Earthquakes, landslides, subsidence, infrastructure, floods and the workflow

06 Getting started
Strengths and limits, free data and tools, and a quick check

01

PART 1

Remote sensing and radar basics

Passive and active sensors, radar bands, and fifty years of radar satellites

BASICS

Passive and active sensors

Optical satellites

Passive: use reflected sunlight

- Work only in daylight
- Blocked by cloud, smoke and haze
- Record colour: what the surface is made of
- Like an ordinary camera

Radar satellites (SAR)

Active: send their own microwave pulses

- Work day and night
- See through cloud, smoke and most rain
- Record roughness, moisture and shape, plus the echo’s exact phase
- Like a flash camera that also measures distance

For hazards this matters: storms, floods and eruptions come with cloud, and monsoon regions can stay cloudy for months.

BASICS

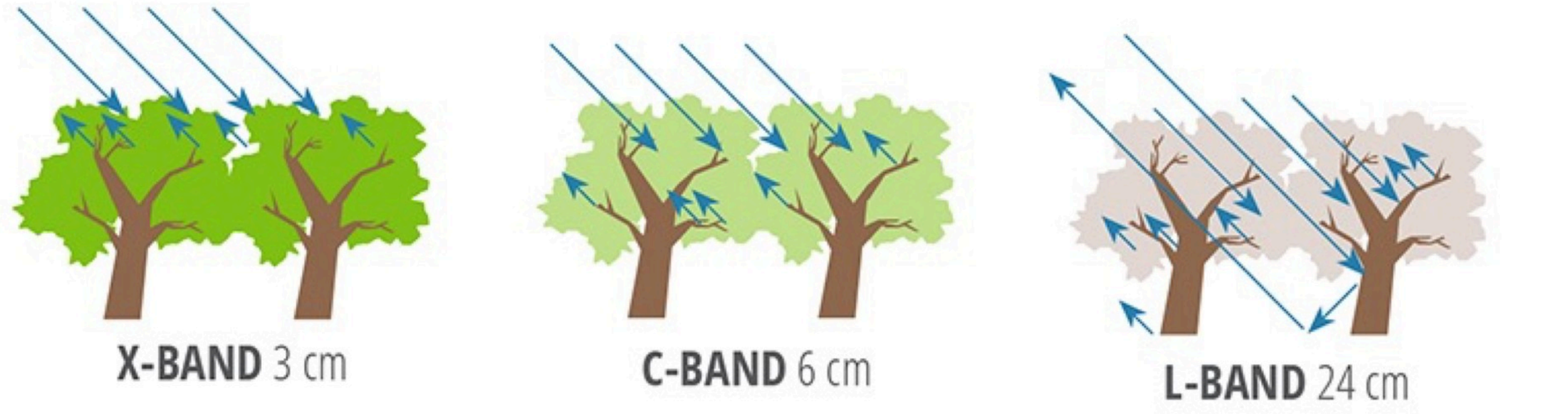
Radar wavelengths and bands

Band	Wavelength	What it sees	Example satellites
X	≈ 3 cm	Fine detail; scatters from surfaces and the tops of vegetation	TerraSAR-X, COSMO-SkyMed, ICEYE, Capella
C	≈ 5.6 cm	All-round workhorse; loses signal in dense vegetation	Sentinel-1, RADARSAT
S	≈ 10 cm	Between C- and L-band	NISAR (ISRO radar), NovaSAR
L	≈ 24 cm	Penetrates vegetation; stays coherent for longer	ALOS-2 / ALOS-4, NISAR, SAOCOM
P	≈ 70 cm	Deep into forests; measures biomass	BIOMASS (ESA, 2025)

Longer wavelengths penetrate further and give fewer, wider fringes; shorter wavelengths show finer detail but lose coherence sooner.

BASICS

Longer waves see deeper



X-band (3 cm) scatters from the leaves at the top of the canopy.

C-band (6 cm) reaches leaves and small branches.

L-band (24 cm) reaches branches, trunks and the ground.

BASICS

Fifty years of radar satellites

1978
SEASAT
NASA’s first civilian imaging radar in orbit (L-band); flew for about 100 days

1991
ERS-1
ESA, C-band. Its data gave the first earthquake interferogram: Landers, 1992

1995
RADARSAT-1
Canada’s first radar satellite (C-band), for operational imaging

2000
SRTM
Space Shuttle mission that mapped near-global topography by InSAR in 11 days

2006
ALOS
Japan’s L-band radar: better results over vegetation and landslides

2007
TerraSAR-X, COSMO-SkyMed
X-band satellites with metre-scale resolution

2014
Sentinel-1
ESA and EU Copernicus, C-band: free and open data worldwide every 6–12 days

2025
NISAR and BIOMASS
NASA–ISRO L- and S-band mission; ESA’s P-band forest mission

02

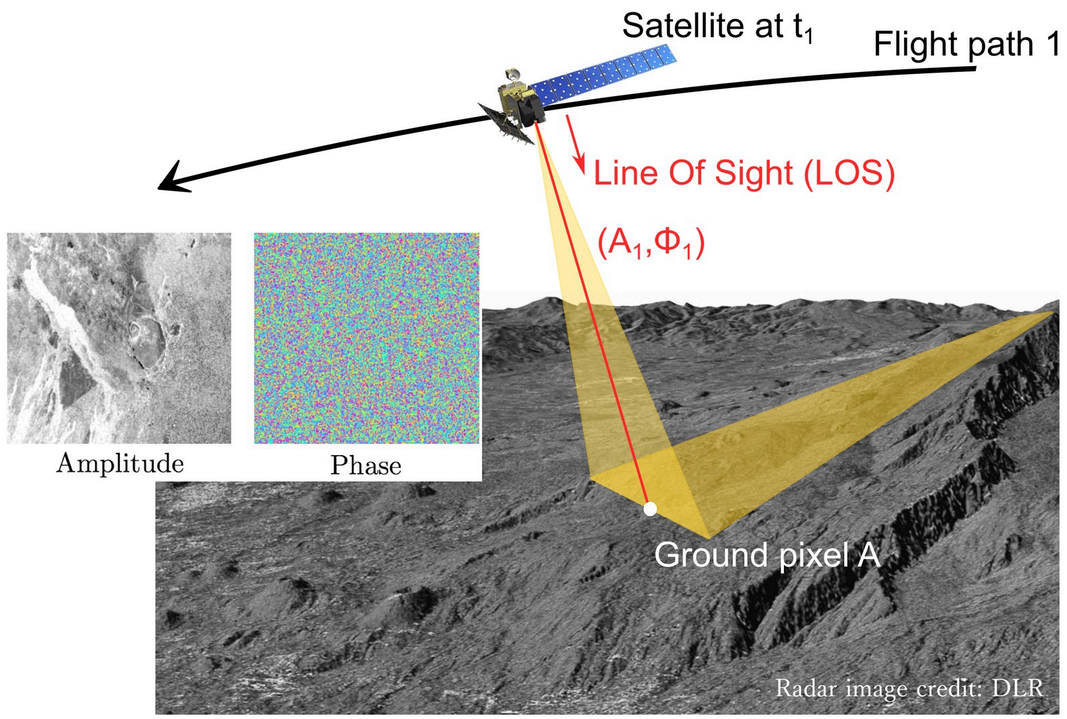
PART 2

What is SAR?

How a radar satellite takes a picture, what the picture shows, and its distortions

SAR

How a radar satellite takes a picture



Radar image credit: DLR

It looks **sideways**, not straight down, from about 700 km up.

It sends more than a thousand microwave pulses every second and times each echo: the delay gives the **distance** (range).

Each pixel records the echo's **strength** (amplitude) and its **phase**, the exact point in the wave cycle when it returns.

Amplitude makes the picture. Phase alone looks random, but compared between two dates it measures motion.

SAR

The “synthetic aperture” trick

A sharp picture from 700–900 km away would need an antenna kilometres long. Instead the satellite records echoes all along its path, and a computer combines them as if they came from one giant antenna.

10 m
Real antenna on the satellite

→

≈ 18 km
Length of ground it lights up as it flies past

→

≈ 36 km
Equivalent “synthetic” antenna

→

≈ 5 m
Resulting resolution on the ground

Numbers from the example in Lecture 1 (L-band, 10 m antenna, 866 km range).

SAR

What a SAR image shows

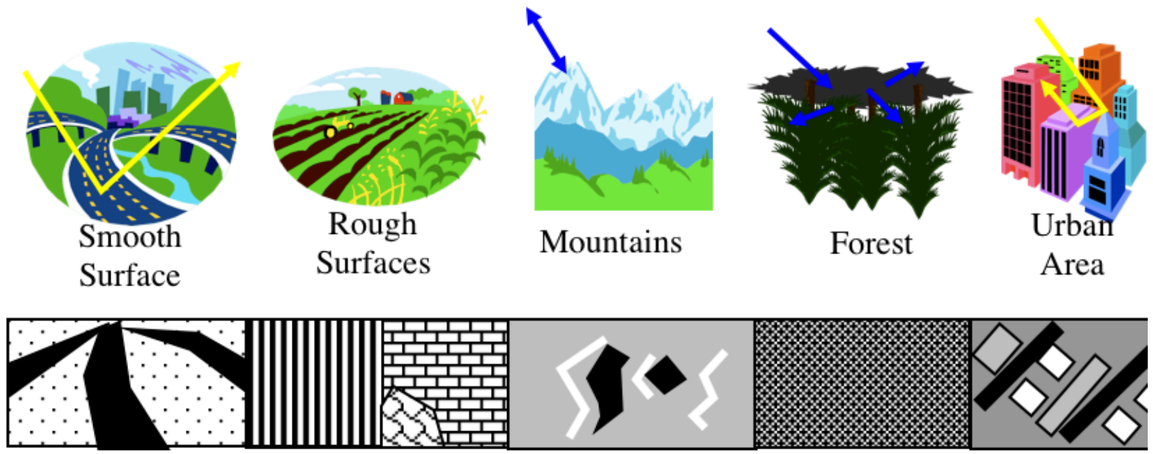


Figure from Lecture 1 (EarthScope short course).

Dark

Smooth surfaces such as calm water, tarmac and airport runways reflect the pulse away, like a mirror.

Bright

Rough ground, forests, slopes facing the radar, and buildings, where walls and ground form a corner (“double bounce”).

It changes

Wet soil is brighter than dry soil; wind roughens water; floods, harvests and snow all change the picture.

SAR

Side-looking geometry distorts mountains

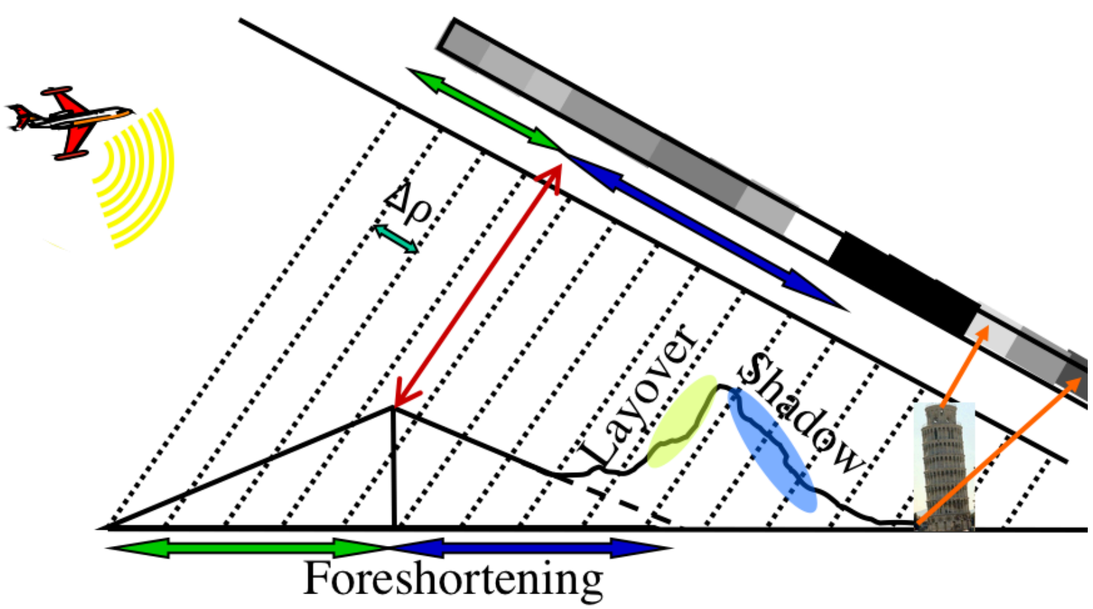


Figure from Lecture 1 (EarthScope short course).

Foreshortening

Slopes facing the radar look shorter and brighter than they are.

Layover

On very steep slopes the top echoes before the bottom, so the slope folds over.

Shadow

Slopes facing away can be completely hidden: no data at all.

03

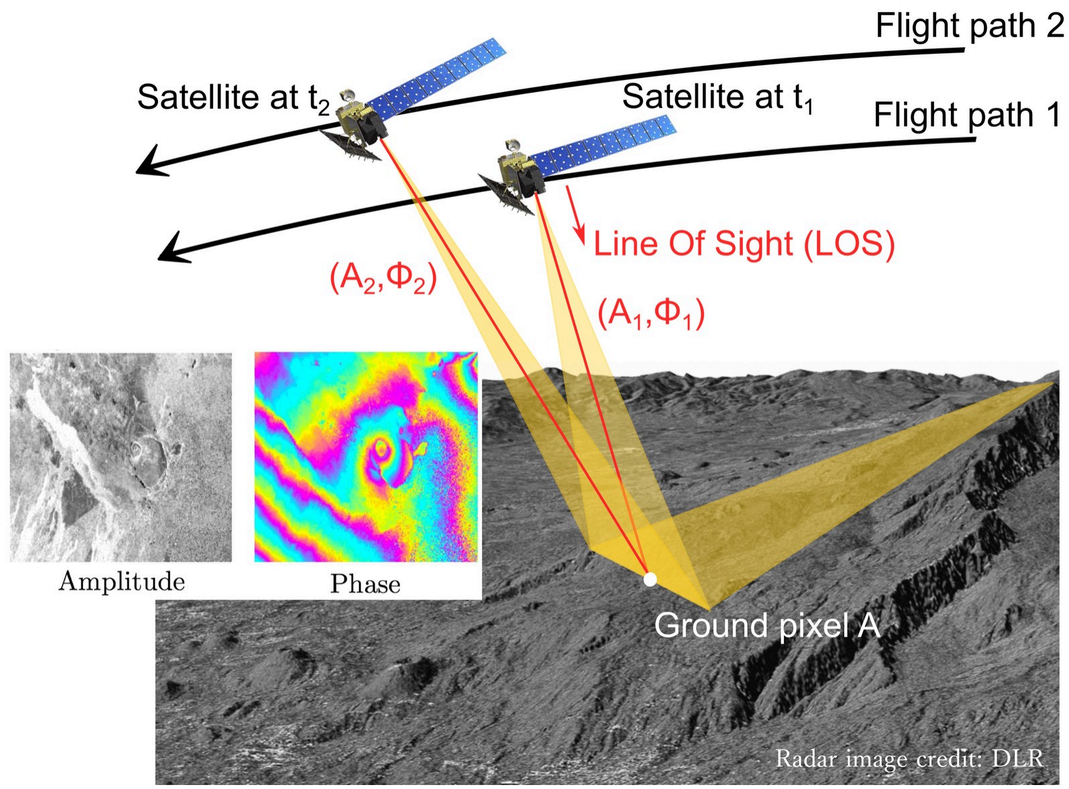
PART 3

What is InSAR?

Phase, fringes, line of sight, time series and coherence

INSAR

Two radar images, one measurement of change

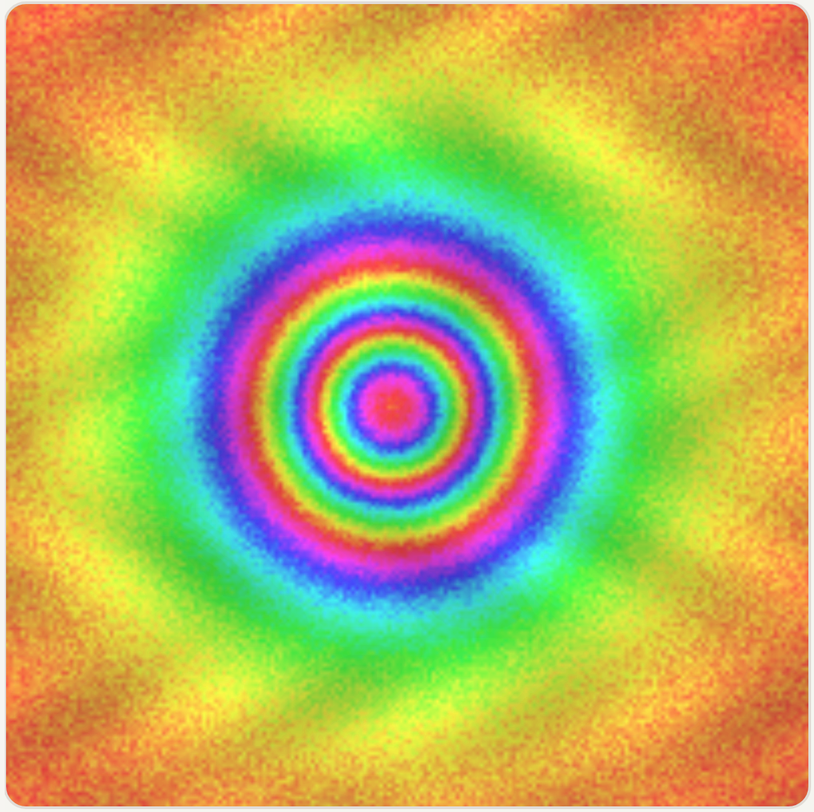


Radar image credit: DLR

- Image the same ground twice from (almost) the same orbit, days or months apart.
- If the ground moved toward or away from the satellite, the echo's phase shifts.
- Subtract the two phases: the result, an **interferogram**, shows coloured bands called **fringes**.
- Each fringe is half a wavelength of motion along the line of sight: **2.8 cm** for Sentinel-1.

INSAR

Watch fringes form



Uplift at the centre

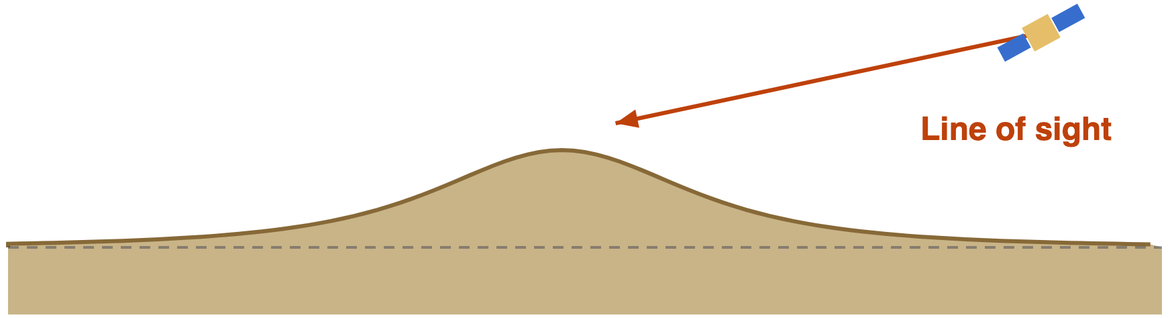
8.5 cm = **3.0 fringes**



0 cm 2.8 cm

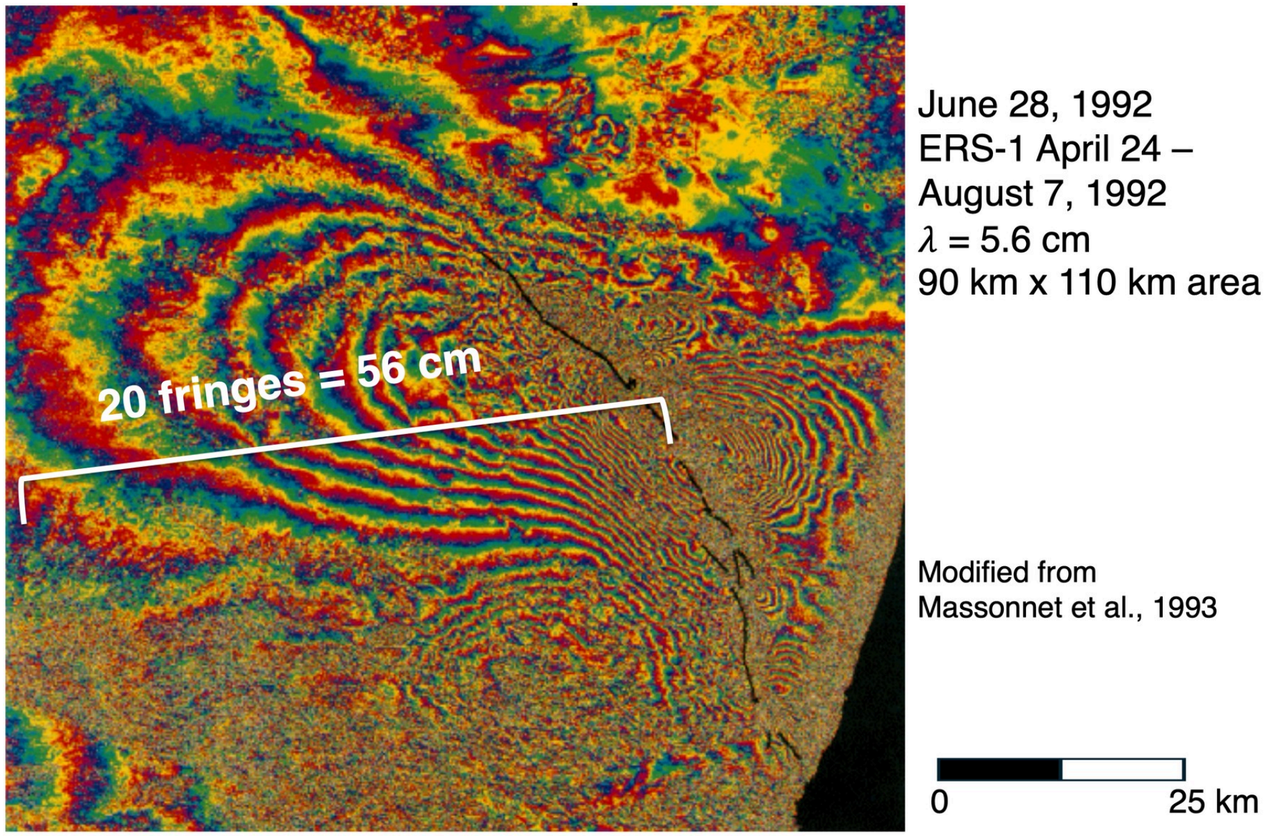
One full colour cycle = λ/2 = 2.8 cm of motion toward the satellite

Cross-section through the volcano (uplift exaggerated)



INSAR

Reading a real interferogram: Landers, 1992



Count the fringes from the edge toward the fault: about 20.

20 × 2.8 cm = **56 cm** of ground motion along the line of sight.

This was the first earthquake ever mapped by InSAR (Massonnet et al., 1993, using ERS-1 C-band data).

Close to the fault, fringes become too dense to count: the ground broke and moved too much.

INSAR

InSAR sees motion in one direction only

A single interferogram measures only the part of the motion along the radar’s line of sight, the slanted direction from the ground to the satellite.

High

Sensitivity to vertical motion (uplift and subsidence)

Medium

Sensitivity to east–west motion

Very low

Sensitivity to north–south motion: the satellites fly roughly north–south

Ascending pass

Satellite moving north, looking east (for right-looking radars such as Sentinel-1).

Descending pass

Satellite moving south, looking west. Combining both separates vertical from east–west motion.

INSAR

From one interferogram to a time series

1

Stack

Collect dozens to hundreds of images of the same area over months or years.

2

Process

Build many interferograms and remove noise from the atmosphere, orbits and topography.

3

Result

A velocity map in mm per year, and a displacement history for every pixel.

Persistent Scatterers (PS-InSAR)

Uses stable, bright points such as buildings, bridges and rock outcrops. Millimetre precision; ideal for infrastructure.

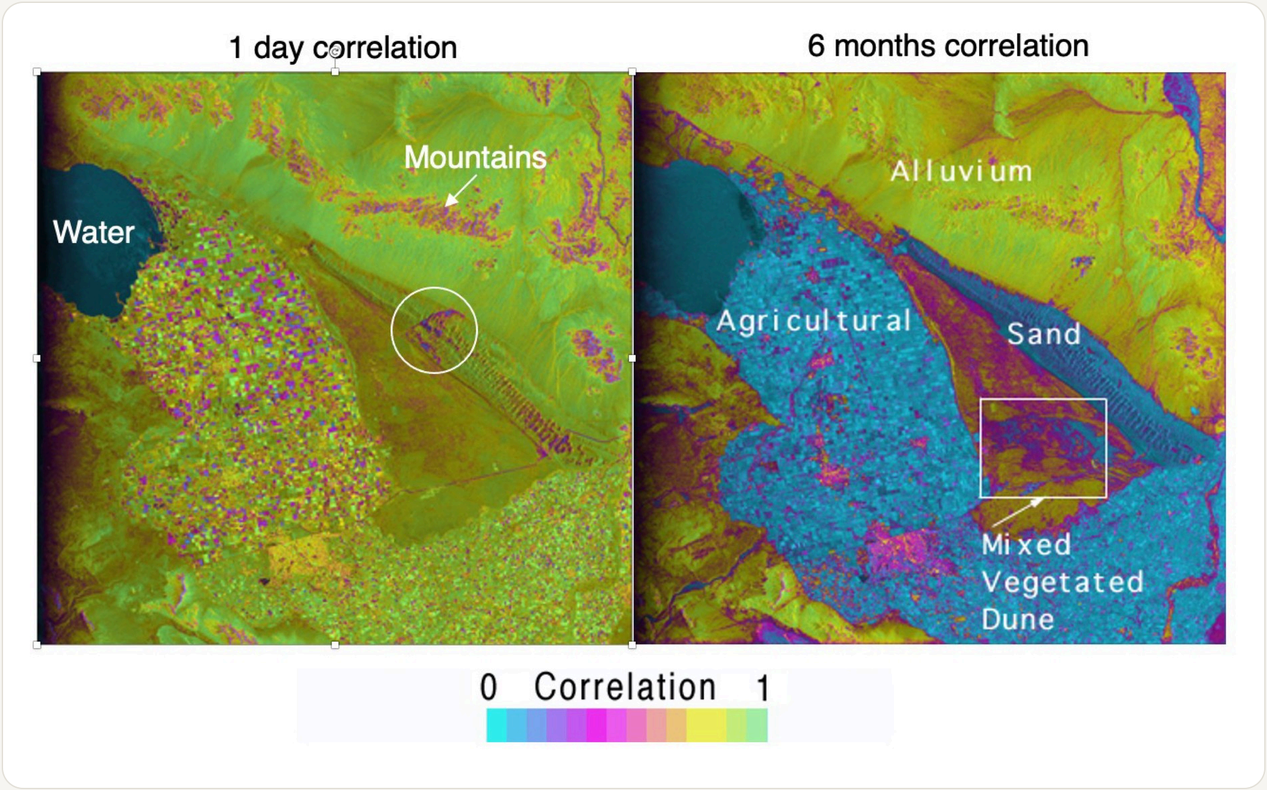
Small Baseline (SBAS)

Uses natural surfaces such as bare soil and fields, averaged over pixels. Good for wide areas and slopes.

Free tools: MintPy, StaMPS, LiCSBAS, ASF HyP3. Ready-made maps: EGMS (all of Europe), COMET-LiCSAR (tectonic and volcanic areas).

INSAR

When InSAR works, and when it doesn’t



Coherence (0 to 1) measures how similar the two images are. High coherence = clean fringes.

High: rock, buildings, bare ground and short time gaps.

Low: water, dense vegetation, snow, ploughed fields and long time gaps (blue on the right map).

Very fast motion: if neighbouring pixels move apart by more than about λ/4 (1.4 cm for Sentinel-1) between passes, the fringes can’t be untangled.

04

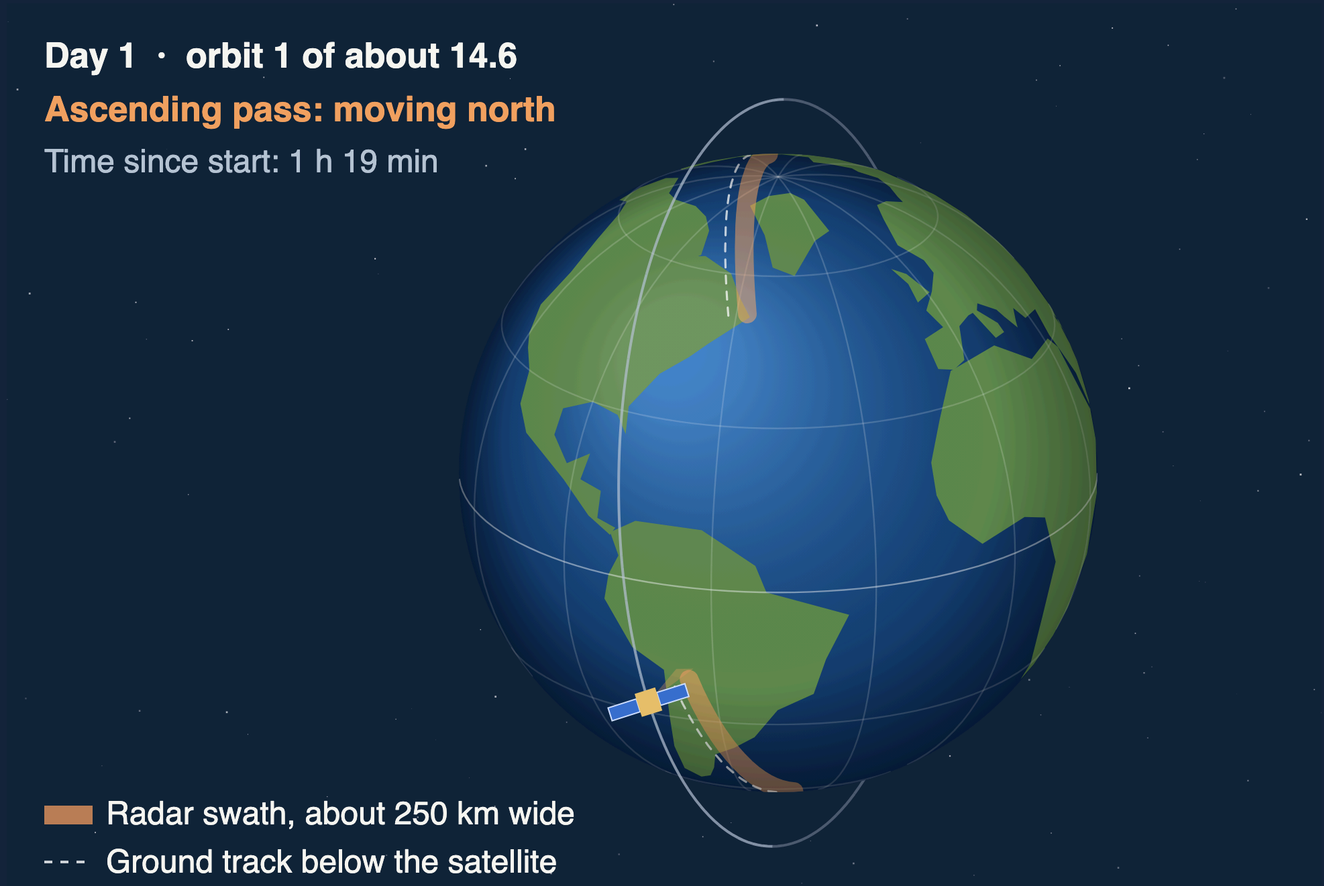
PART 4

Satellites and orbits

How radar satellites circle the Earth, Sentinel-1, NISAR and other missions

SATELLITES

How a radar satellite circles the Earth



- Near-polar orbit about 700 km up, passing close to both poles.
- One orbit takes about 99 minutes: roughly 14.6 orbits a day.
- Earth turns underneath, so each pass images a new strip (orange).
- Going north is an **ascending** pass; going south, a **descending** pass.
- After 12 days (175 orbits) Sentinel-1 flies exactly the same path again. That repeat is what makes InSAR possible.

Animation: one orbit shown in about 5 seconds; satellite height and swath width exaggerated.

SATELLITES

Sentinel-1 in numbers

693 km
Orbit altitude

98.6 min
Time for one orbit, about 14.6 orbits a day

12 days
Exact repeat cycle (6 days when two satellites fly together)

250 km
Swath width in the standard Interferometric Wide mode

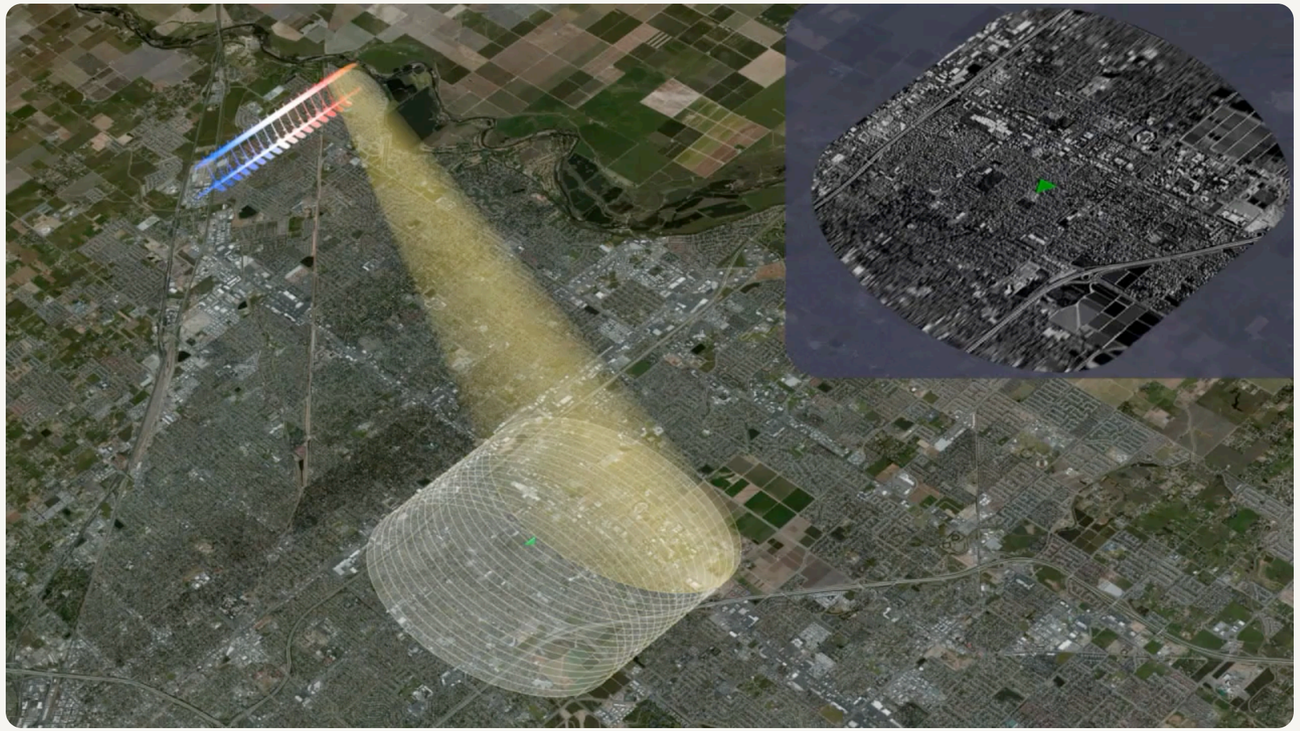
5 × 20 m
Pixel size in range and azimuth

Free
All data open to everyone since 2014 (EU Copernicus)

Sun-synchronous dawn–dusk orbit: it passes over each place at about 6 am and 6 pm local time, on descending and ascending passes.

SATELLITES

NISAR: the NASA–ISRO radar mission



Animation: NISAR imaging the ground (plays when presenting)

- Launched on 30 July 2025 from Sriharikota, India, on a GSLV rocket.
- Two radars: L-band (24 cm, NASA) and S-band (about 10 cm, ISRO).
- A 12 m reflector antenna and a swath more than 240 km wide.
- Maps nearly all of Earth’s land and ice every 12 days.
- Free and open data: landslides, subsidence, earthquakes, glaciers, crops and forests.

SATELLITES

Radar satellites you can use

Mission	Agency	Band	Repeat	Data
Sentinel-1	ESA / EU Copernicus	C	12 days (6 with two)	Free and open
NISAR	NASA and ISRO	L + S	12 days	Free and open
ALOS-2 / ALOS-4	JAXA (Japan)	L	14 days	Mostly commercial; research proposals
RADARSAT Constellation	CSA (Canada)	C	4 days	Restricted
SAOCOM	CONAE (Argentina)	L	8 days (pair)	On request
TerraSAR-X / TanDEM-X	DLR and Airbus	X	11 days	Commercial; research proposals
COSMO-SkyMed	ASI (Italy)	X	16 days, less as a constellation	Commercial; research proposals
ICEYE, Capella	Commercial companies	X	Daily or better	Commercial; some open data

05

PART 5

Hazards and engineering

Earthquakes, landslides, subsidence, infrastructure, floods and the workflow

APPLICATIONS

What InSAR does for hazards and engineering

Earthquakes

Map how far and where the ground moved; locate the fault and estimate its slip within days.

Volcanoes

Detect swelling as magma collects, sometimes weeks to years before an eruption.

Landslides

Find slow-moving slopes, measure mm–cm per year of creep, and spot acceleration before failure.

Land subsidence

Measure sinking from groundwater pumping, mining, tunnelling and soft soils.

Infrastructure

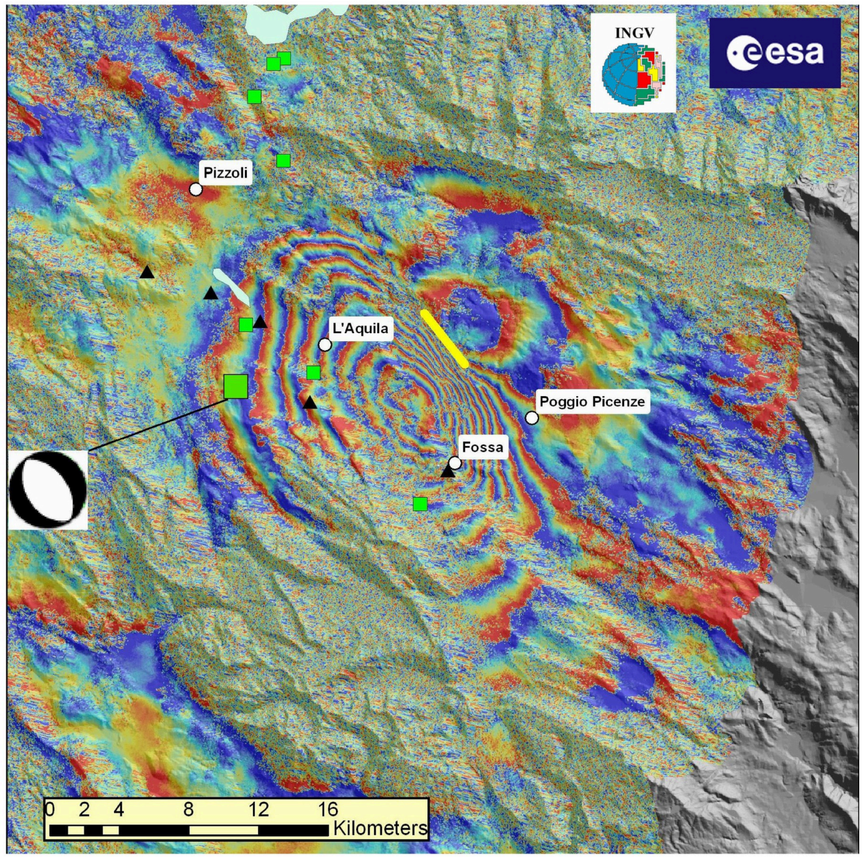
Monitor settlement of buildings, bridges, dams, embankments, railways and pipelines.

Floods and damage

Map flood extent through cloud, and find damaged areas where coherence drops.

APPLICATIONS

Earthquakes: L’Aquila, Italy, 2009



Magnitude 6.3, 6 April 2009, central Italy.

Concentric fringes mark a bowl of ground that dropped during the earthquake. One C-band fringe = 2.8 cm.

Try it: count the fringes from the edge of the bowl to its centre and multiply by 2.8 cm.

Uses: locate the fault that moved, model its slip, and guide damage surveys and rebuilding.

Envisat interferogram interpreted by INGV; image ESA. From the EarthScope InSAR theory slides.

APPLICATIONS

Landslides and unstable slopes

What InSAR gives you

- An inventory of active slopes across whole regions
- Velocity in mm to cm per year
- Seasonal speed-ups after the monsoon or snowmelt
- Acceleration: a warning sign before failure

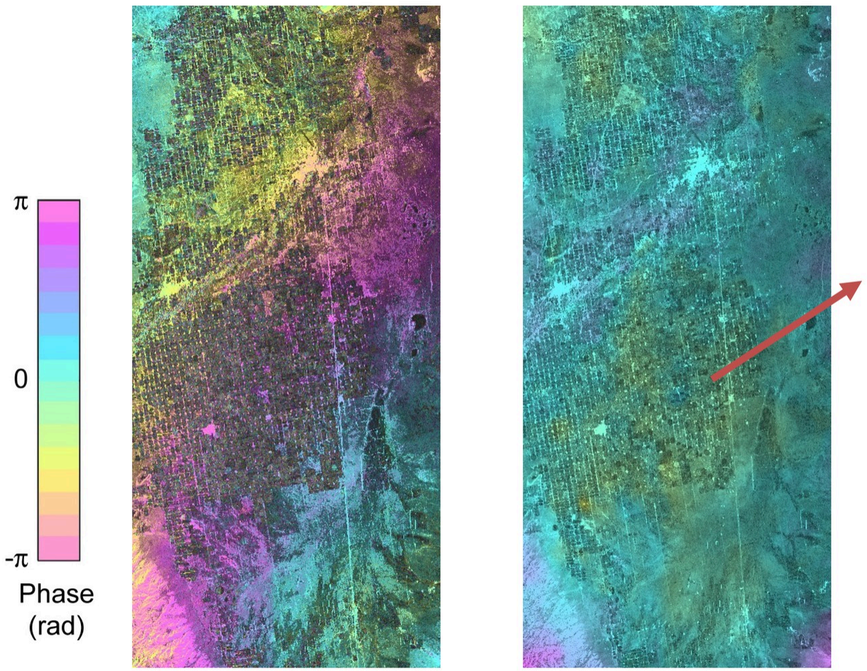
Watch out for

- Steep slopes in layover or shadow
- Vegetation and snow: low coherence (prefer L-band)
- Slopes moving north or south: hard to see
- Fast slides (metres per day): too fast, so use optical images or ground sensors

Engineering use: prioritise slopes for site investigation, design drainage and stabilisation, and check that remedial works are actually working.

APPLICATIONS

Land subsidence: groundwater, mining and cities



An L-band interferogram before (left) and after (right) the orbital ramp removal.

- Pumping groundwater compacts clay layers: whole cities can sink by centimetres a year.
- Recharge can lift the ground back up, as in the light-yellow area here.
- Mining, tunnelling and the consolidation of soft soils also show up clearly.
- Used for groundwater management, flood-risk planning, and protecting pipelines, railways and foundations.

APPLICATIONS

Infrastructure and engineering monitoring

Buildings and foundations

Settlement and tilt, including damage from nearby excavation.

Bridges and viaducts

Separate seasonal thermal expansion from real movement.

Dams and embankments

Settlement of the crest and stability of reservoir slopes.

Railways and highways

Subsidence and slope movement along whole corridors.

Tunnels and metros

Ground settlement above tunnels during and after construction.

Mines and tailings dams

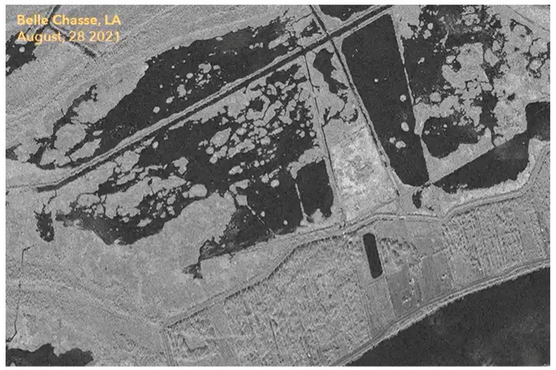
Deformation of waste dumps, pit walls and dams.

PS-InSAR reaches millimetre precision on hard targets; where there are none, small metal **corner reflectors** can be installed as targets.

APPLICATIONS

Floods and damage mapping

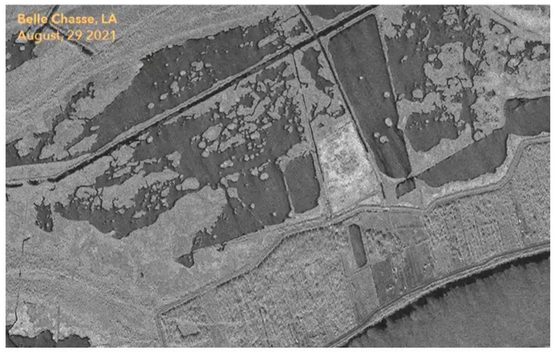
Day 1– normal conditions



Day 3 – flooded and possibly windy



Day 2 – windy (high reflectivity on water)



Day 4 – less flooded, not windy



Calm water is dark in radar images, so floods stand out, even under storm clouds.

Comparing before and after images maps a flood within hours of a satellite pass.

Loss of coherence between images reveals damaged buildings after earthquakes or storms.

Example: Belle Chasse, Louisiana, 28–31 August 2021, as Hurricane Ida passed.

IN PRACTICE

From satellite data to engineering decisions

1
Data
Download Sentinel-1 or NISAR scenes (ASF, Copernicus)

→

2
Interferograms
Process with ISCE2, SNAP or ASF HyP3

→

3
Time series
Velocity maps with MintPy, LiCSBAS or StaMPS

→

4
Interpret
In QGIS, with geology, slopes, land use and assets

→

5
Validate
GNSS, levelling, inclinometers and field visits

InSAR tells you **where** the ground is moving and **how fast**. The engineer works out **why**, and **what to do** about it.

06

PART 6

Getting started

Strengths and limits, free data and tools, and a quick check

GETTING STARTED

Strengths and limitations

Strengths

- Day and night, through cloud
- Thousands of km² in one image
- Millimetre to centimetre precision
- Archive back to 1992: look into the past
- Free data from Sentinel-1 and NISAR
- No instruments needed on site

Limitations

- Measures only line-of-sight motion
- Nearly blind to north–south motion
- Loses coherence on vegetation, snow and water
- Struggles with fast or large motion
- Atmosphere can mimic deformation
- Layover and shadow in steep terrain
- Always needs ground truth

GETTING STARTED

Free data and tools

ASF Data Search

search.asf.alaska.edu

Sentinel-1, NISAR and ALOS archives; on-demand interferograms with HyP3

Copernicus Data Space

dataspace.copernicus.eu

Official Sentinel-1 data access and browser

COMET-LiCSAR

comet.nerc.ac.uk/comet-lics-portal

Ready-made interferograms for tectonic and volcanic areas

EGMS

egms.land.copernicus.eu

Ground-motion maps for all of Europe, ready to view

ESA SNAP

step.esa.int

Free desktop software to make interferograms

MintPy

github.com/insarlab/MintPy

Open-source InSAR time-series analysis in Python

CHECK YOUR UNDERSTANDING

Three quick questions

Question 1

Why can a radar satellite image a flood during a storm, when an optical satellite can’t?

Question 2

A Sentinel-1 interferogram shows 5 fringes across a landslide. About how much did it move along the line of sight?

Question 3

Why do we combine ascending and descending passes?

ANSWERS

Three quick answers

Answer 1

Radar sends its own microwave pulses, centimetres long, which pass through cloud and work at night.

Answer 2

5 × 2.8 cm = 14 cm along the line of sight.

Answer 3

Each pass looks from a different direction; together they separate vertical motion from east–west motion.

RECAP

Key ideas

01 SAR is an active radar that images day and night, through cloud.

02 Each pixel holds an amplitude and a phase; phase differences measure changes in distance.

03 One fringe is λ/2 of line-of-sight motion: 2.8 cm for Sentinel-1.

04 Repeat orbits every few days and free data make routine hazard monitoring possible.

05 InSAR shows where and how fast; engineers combine it with ground data to decide why and what to do.

COMING UP

The rest of this module

Lecture 1

SAR principles and phenomenology: how radar forms an image, resolution, Doppler, scattering and speckle.

Lecture 2

Geometric aspects of InSAR: phase, baselines, topography, deformation and error sources, with quizzes.

Lecture 3

InSAR theory with real data: critical baseline, unwrapping, atmosphere, correlation and wavelength.

Thank you. Questions?

Figures reused from the EarthScope InSAR short-course materials (github.com/parosen/Geo-SInC): InSAR theory slides by M. Zebker and A. Chen (after P. A. Rosen and S. Hensley) and the SAR lecture by P. A. Rosen; image credits DLR, ESA, INGV and Massonnet et al. (1993). NISAR animation: NISAR mission. Orbit and fringe animations simulated for this lecture.